CHAPTER 9
PRESSURE

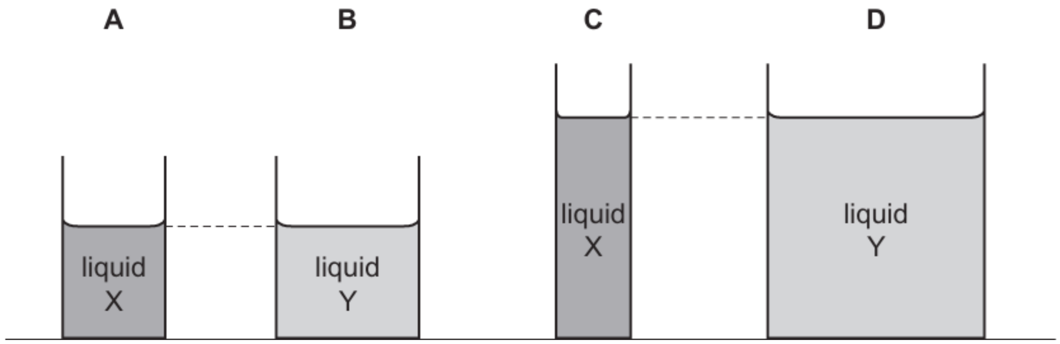
1.8	**Pressure**
1	Define pressure as force per unit area; recall and use the equation $\text{pressure} = \dfrac{\text{force}}{\text{area}}$ $p = \dfrac{F}{A}$
2	Describe how pressure varies with force and area in the context of everyday examples
3	State that the pressure at a surface produces a force in a direction at right angles to the surface and describe an experiment to show this
4	Describe how the height of a liquid column in a liquid barometer may be used to determine the atmospheric pressure
5	Describe, quantitatively, how the pressure beneath the surface of a liquid changes with depth and density of the liquid
6	Recall and use the equation for the change in pressure beneath the surface of a liquid change in pressure = density × gravitational field strength × change in height $\Delta p = \rho g \Delta h$

1. O/N/23/11/Q12

Two liquids, X and Y, are poured into beakers of different sizes.

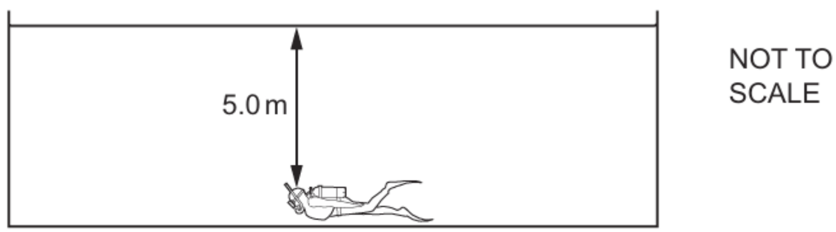
Liquid X has a density of $1000\,kg/m^3$ and liquid Y has a density of $900\,kg/m^3$.

On the base of which beaker is the pressure greatest?



2. O/N/23/11/Q13

A swimmer is at the bottom of a pool. The density of the water in the pool is $1000\,kg/m^3$ and the gravitational field strength is $9.8\,N/kg$.



Which expression gives the pressure exerted on the swimmer due to the water?

A $\frac{5.0 \times 1000}{9.8}$ Pa

B $5.0 \times 1000 \times 9.8$ Pa

C $\frac{1000}{5.0}$ Pa

D $\frac{1000 \times 9.8}{5.0}$ Pa

3. O/N/23/12/Q16

A man finds it difficult to hammer a wooden post into the ground.

How could he make the post go in more easily?

A Make the bottom of the post pointed.

B Make the top of the post pointed.

C Use a longer post.

D Use a wider post.

4. M/J/23/11/Q13

An African elephant has a mass of 6000 kg and each of its four feet has an area of $0.20\,m^2$. The gravitational field strength g is 9.8 N / kg.

What is the average pressure due to the elephant when it is standing stationary on all four feet on level ground?

A 7.5 kPa **B** 30 kPa **C** 74 kPa **D** 290 kPa

5. O/N/22/11/Q13

Each tyre of a car has an area of $100\,cm^2$ in contact with the ground.

The car has a mass of 1600 kg. The weight of the car is equally distributed amongst the four tyres.

The gravitational field strength g is 10 N / kg.

What is the pressure exerted on the ground?

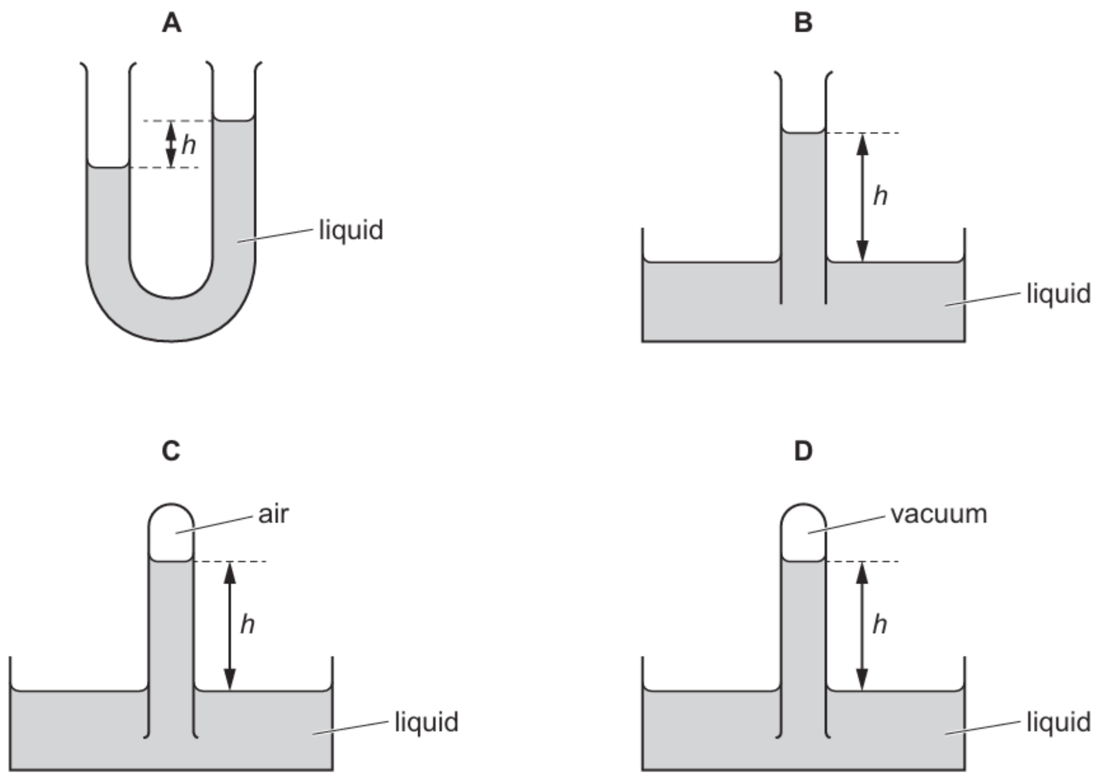
A $4.0\,N/cm^2$ **B** $16\,N/cm^2$ **C** $40\,N/cm^2$ **D** $160\,N/cm^2$

6. M/J/22/12/Q11

A barometer is an instrument used to measure atmospheric pressure.

In one type of barometer the height of a liquid in a tube is measured.

In which diagram does the height *h* represent atmospheric pressure?

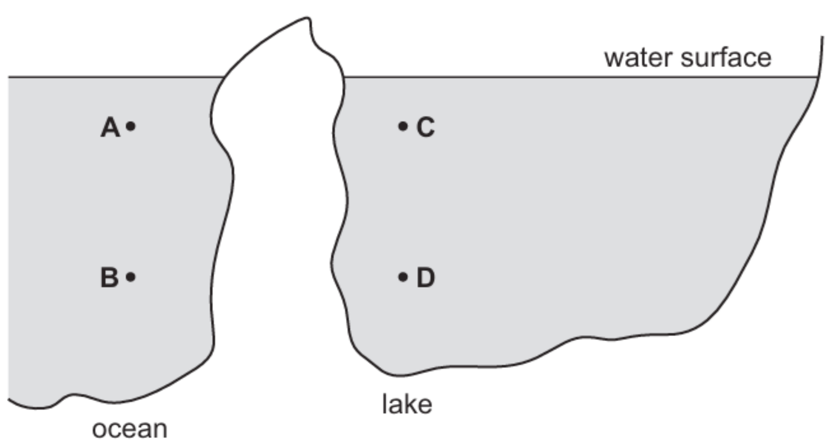


7. O/N/21/11/Q8

An underwater diver moves from the ocean to a fresh water lake.

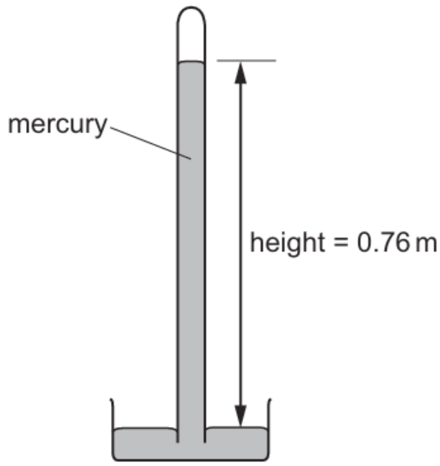
The density of water in the lake is less than in the ocean.

In which position does the diver experience the smallest pressure due to the water?



8. O/N/21/11/Q10

When the pressure exerted by the atmosphere is 0.10 MPa, the height of the mercury in a barometer is 0.76 m.



What is the pressure exerted by the atmosphere when the height of the mercury in the simple barometer is 0.57 m?

A 0.025 MPa **B** 0.075 MPa **C** 0.10 MPa **D** 0.13 MPa

9. M/J/21/11/Q11

Which expression for pressure is correct?

A force × area

B force ÷ area

C mass × area

D mass ÷ area

10. M/J/21/11/Q12

At a depth d in sea-water, the total pressure experienced by a diver is $2P$, where P is atmospheric pressure.

At which depth is the pressure $4P$?

A $1.5d$ **B** $2d$ **C** $3d$ **D** $4d$

11. M/J/21/12/Q12

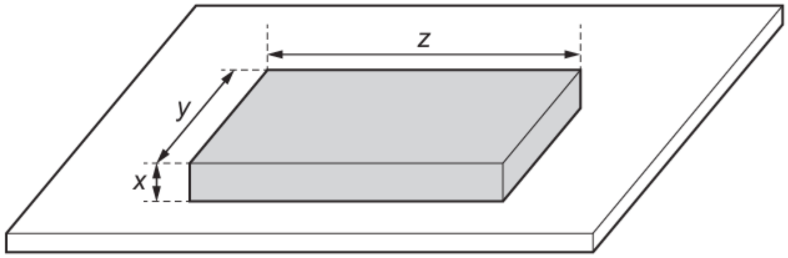
An object is placed at a depth d below the surface of a liquid of density ρ.

On which expression does the pressure on the object depend?

A $d + \rho$ **B** $d \times \rho$ **C** $d \div \rho$ **D** $\rho \div$ d

12. M/J/21/12/Q13

The diagram shows the dimensions of a box of mass M and weight W at rest on a table.

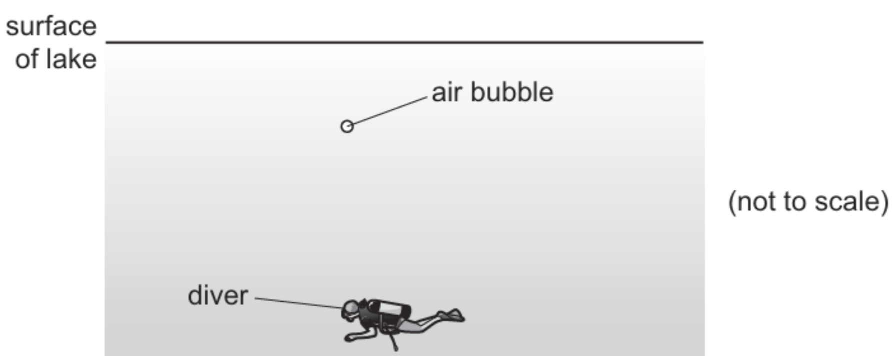


What is the pressure on the table due to the box?

A $\dfrac{M}{xyz}$ **B** $\dfrac{M}{yz}$ **C** $\dfrac{W}{xyz}$ **D** $\dfrac{W}{yz}$

13. O/N/20/12/Q12

The diagram shows a diver 20 m below the surface of a lake. The total pressure at this depth is 3.0×10^5 Pa.



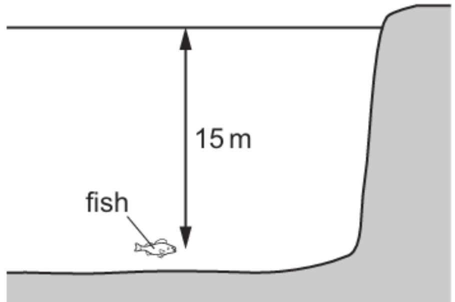
An air bubble has a volume of $0.60\,\text{cm}^3$ as it leaves the diver. It rises to the surface of the lake where the pressure is 1.0×10^5 Pa. The temperature of the air in the bubble remains constant.

What is the volume of the air bubble at the surface of the lake?

A $0.20\,\text{cm}^3$ **B** $0.60\,\text{cm}^3$ **C** $1.8\,\text{cm}^3$ **D** $2.4\,\text{cm}^3$

14. M/J/20/11/Q14

A fish is swimming 15 m below the surface of a lake, as shown.



The density of the water is $1000\,kg/m^3$.

Atmospheric pressure is 100 000 Pa.

The acceleration of free fall g is $10\,m/s^2$.

What is the total pressure on the fish?

A 50 000 Pa **B** 120 000 Pa **C** 150 000 Pa **D** 250 000 Pa

15. M/J/20/12/Q14

An object is placed at different depths in liquids of different densities.

In which liquid and at what depth is the pressure on the object the greatest?

	density of liquid / kg/m^3	depth / m
A	1000	4.0
B	1000	10
C	1200	4.0
D	1200	10

16. O/N/19/11/Q15

The list gives the symbols for some physical quantities.

acceleration of free-fall g

atmospheric pressure P

density of water ρ

depth of water h

Which quantities from the list are used to calculate the total pressure at the bottom of a lake?

A g, h, P and ρ

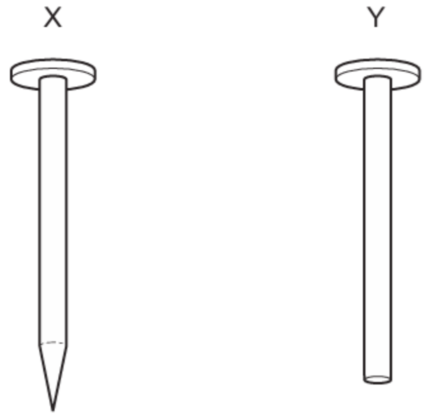
B g, h and P only

C h, P and ρ only

D g, h and ρ only

17. O/N/19/11/Q14

Nail X is tapped into wood. Nail Y is tapped into the same wood using the same force.

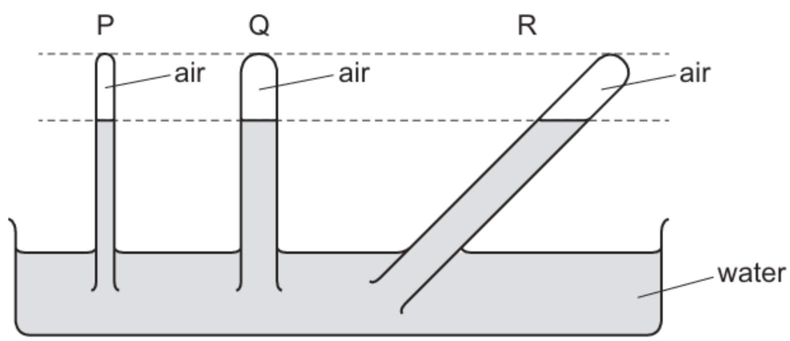


Which nail moves furthest into the wood and why?

	nail	why
A	X	X produces greater pressure on the wood
B	X	X produces smaller pressure on the wood
C	Y	Y produces greater pressure on the wood
D	Y	Y produces smaller pressure on the wood

18. O/N/19/12/Q12

The diagram shows three tubes P, Q and R. Each tube contains air trapped by a water column.



Which statement is correct?

A The pressure of the trapped air is equal in tubes P, Q and R.

B The pressure of the trapped air is greatest in tube Q.

C The pressure of the trapped air is greatest in tube R.

D The pressure of the trapped air is greatest in tube P.

19. M/J/19/11/Q11

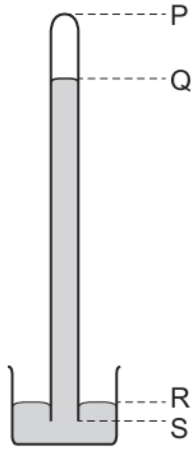
A garden table weighs 60 N and has a top surface of area $2.0\,m^2$. It is raining and the rain produces a pressure of $4.0\,N/m^2$ on the table.

Ignoring the pressure of the atmosphere, what is the force exerted by the table on the ground?

A 52 N **B** 58 N **C** 62 N **D** 68 N

20. O/N/18/11/Q11

A long tube, full of mercury, is inverted in a small dish of mercury.



The mercury level in the tube falls, leaving a vacuum at the top.

When the atmospheric pressure decreases, which length decreases?

A PQ **B** PS **C** QR **D** RS

21. O/N/18/12/Q13

Different liquids are poured into four different containers.

In which container does the liquid produce the greatest pressure at the bottom of the container?

	area of base / cm^2	$\frac{\text{density of liquid}}{g/cm^3}$	depth of liquid / cm
A	10	1.3	50
B	20	0.80	80
C	40	1.0	60
D	50	0.92	75

22. O/N/18/12/Q14

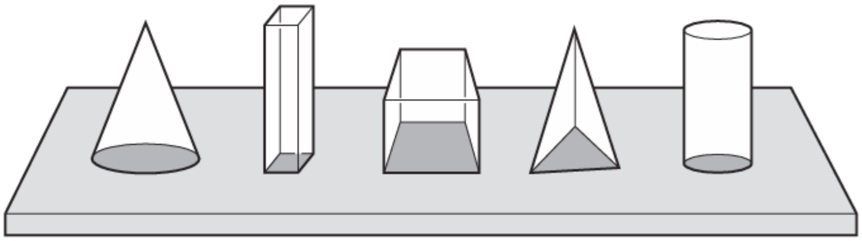
A rectangular block of metal has weight 6.0 N and measures 3.0 cm × 4.0 cm × 5.0 cm.

What is the smallest pressure that it can exert when resting on a horizontal surface?

A 0.10 N/cm^2 **B** 0.30 N/cm^2 **C** 0.40 N/cm^2 **D** 0.50 N/cm^2

23. M/J/18/11/Q14

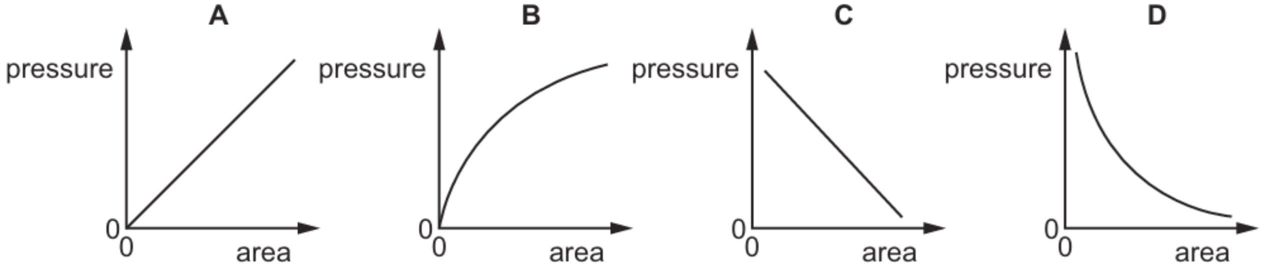
Five blocks have the same mass but different base areas. They all rest on a horizontal table.



A graph is plotted to show the relationship between the pressure exerted on the table and the base area of the block.

Which graph shows this relationship?

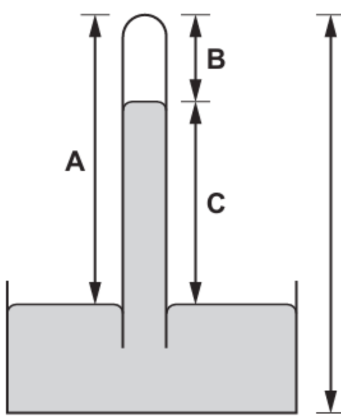
A **B** **C** **D**



24. M/J/18/11/Q16

The diagram shows a simple mercury barometer.

Which height is a measure of the atmospheric pressure?



25. M/J/18/12/Q18

A sealed packet containing air and a snack is purchased at an airport. The sealed packet is taken on board an aircraft. During the flight the packet becomes larger.

What causes the packet to become larger?

A The density of the air inside the packet increases.

B The mass of the packet increases.

C The pressure of the air outside the packet decreases.

D The volume of the air inside the packet decreases.

26. M/J/18/12/Q19

A sealed container of gas is heated and the pressure inside increases.

What happens to the molecules of the gas to cause this increase in pressure?

A Their kinetic energy decreases.

B They become heavier.

C They expand.

D They hit the container more frequently.

27. O/N/17/12/Q12

An extremely large increase in pressure is needed to compress a liquid. A gas can be compressed by a much smaller increase in pressure.

Which statement explains this?

A Molecules repel each other very strongly when very close.

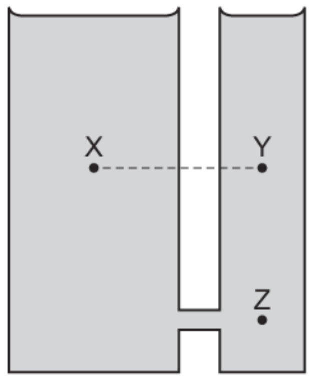
B The attractive forces between molecules are small at large distances.

C The molecules in the gas collide with the walls of their container and rebound.

D The molecules of a liquid are constantly moving at random.

28. M/J/17/11/Q16

Two cylindrical vessels are joined together and filled with water as shown.



How does the pressure at point X compare to the pressure at points Y and Z?

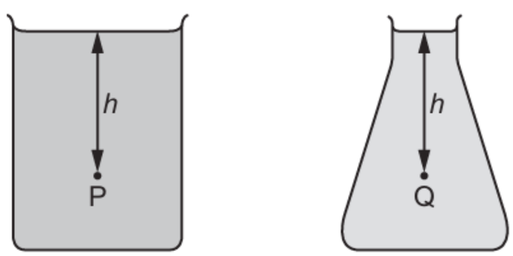
	compared to Y	compared to Z
A	pressure at X is higher	pressure at X is lower
B	pressure at X is higher	pressure at X is the same
C	pressure at X is the same	pressure at X is lower
D	pressure at X is the same	pressure at X is the same

29. M/J/17/12/Q18

Two glass containers filled with different liquids are placed next to each other.

Point P is a distance *h* below the surface of the liquid in one container.

Point Q is a distance *h* below the surface of the liquid in the other container.



Why is the pressure at P different from the pressure at Q?

A The atmospheric pressure is different at P.

B The densities of the liquids are different.

C The gravitational field strength is different at P.

D The shapes of the containers are different.

30. O/N/16/11/Q12

The pressure of the atmosphere is 100 000 Pa.

What force does the atmosphere exert on the upper surface of a pond of surface area 20 m^2?

A 5000 N **B** 100 000 N **C** 100 020 N **D** 2 000 000 N

31. O/N/16/11/Q13

Oil of density 8.5×10^2 kg / m^3 is stored in a large tank.

The gravitational field strength g is 10 N / kg.

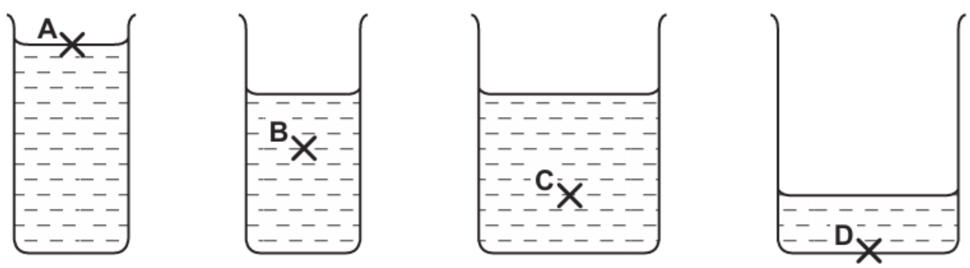
What is the pressure due to the oil 6.0 m below its surface?

A 51 Pa **B** 510 Pa **C** 5100 Pa **D** 51 000 Pa

32. M/J/16/11/Q15

Four beakers contain the same liquid.

At which point is the pressure the greatest?



33. M/J/16/11/Q16

Water of depth 10 m exerts a pressure equal to atmospheric pressure.

An air bubble rises to the surface of a lake which is 20 m deep. When the bubble reaches the surface, its volume is 6.0 cm^3.

What is the volume of the air bubble at the bottom of the lake?

A 2.0 cm^3 **B** 3.0 cm^3 **C** 12 cm^3 **D** 18 cm^3

34. M/J/16/12/Q11

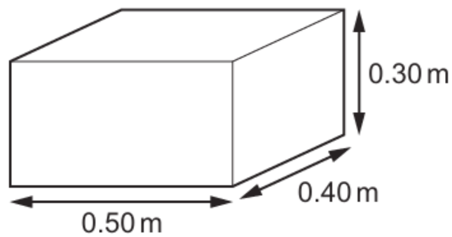
A block of weight W rests on a side of area A. The gravitational field strength is g.

What is the pressure exerted on the ground due to the block?

A WA **B** $\dfrac{W}{A}$ **C** $\dfrac{WA}{g}$ **D** $\dfrac{W}{g}$

35. O/N/15/11/Q11

A block of weight 900 N has rectangular faces. The diagram shows the lengths of the sides.



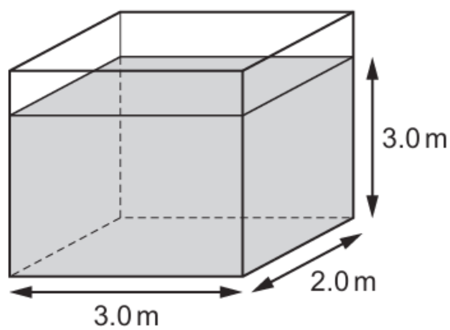
The block can rest on any of its faces.

What is the minimum pressure that the block can exert on the ground when resting on one of its faces?

A 900 Pa **B** 4500 Pa **C** 6000 Pa **D** 7500 Pa

36. O/N/15/12/Q13

The base of a rectangular storage tank is 2.0 m by 3.0 m. The tank is filled with paraffin to a depth of 3.0 m.



The density of paraffin is 800 kg / m^3 and the gravitational field strength is 10 N / kg.

What is the pressure at the bottom of the tank due to the paraffin?

A 2400 Pa **B** 14 400 Pa **C** 24 000 Pa **D** 144 000 Pa

37. O/N/15/12/Q15

A piston of area 10 cm^2 is pushed slowly into a very large cylinder containing gas at a pressure of 10 N / cm^2. The pressure of the gas remains constant as the piston moves a distance of 0.10 m.

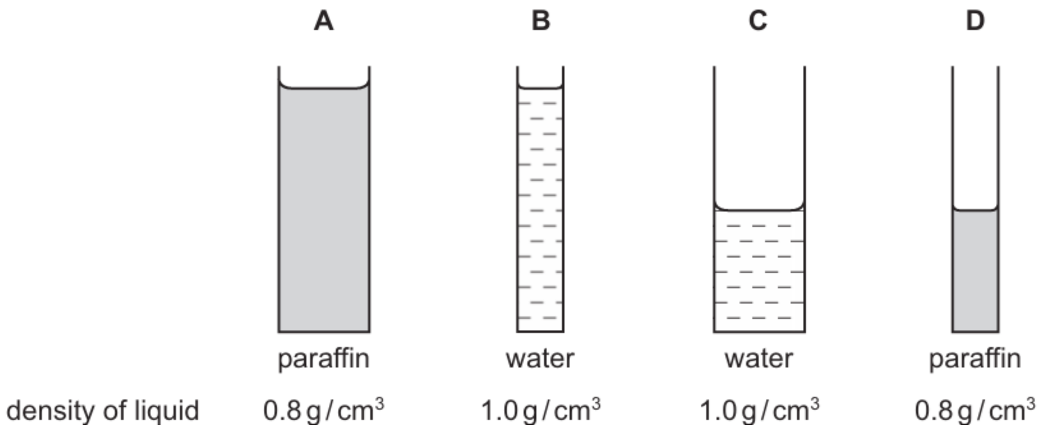
What is the force of the gas on the piston and what is the work done by the piston on the gas?

	force / N	work done / J
A	1.0	0.1
B	1.0	10
C	100	10
D	100	1000

38. M/J/15/11/Q14

The diagrams show liquids in containers.

Which column of liquid exerts the greatest pressure on the base of its container?



39. O/N/14/11/Q10

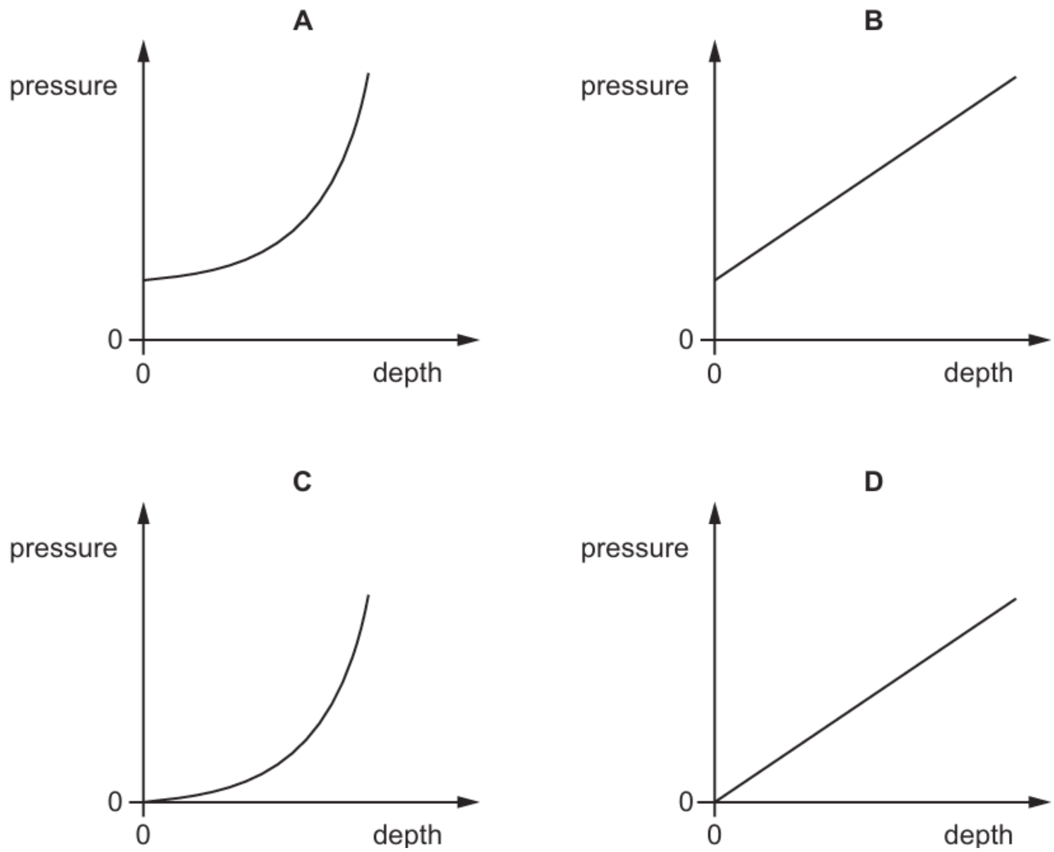
Objects of different weights are placed on a rigid, horizontal surface.

Which row shows the correct pressure acting on the surface?

	weight / N	area in contact / m^2	pressure / Pa
A	10	0.1	1
B	20	0.2	0.01
C	30	0.1	300
D	40	0.2	8

40. M/J/14/12/Q14

Which graph shows the total external pressure acting on a submarine at different depths below the surface of the sea?



41. O/N/13/11/Q9

Which source releases carbon dioxide, a greenhouse gas, when generating electricity?

A fossil fuels

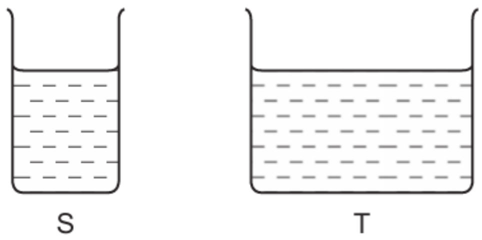
B geothermal

C hydroelectric

D nuclear

42. O/N/13/12/Q10

Two vessels S and T are filled to the same level with the same liquid. The area of the base of S is less than that of T.



Which statement is correct?

A The force on the base of S is greater than the force on the base of T.

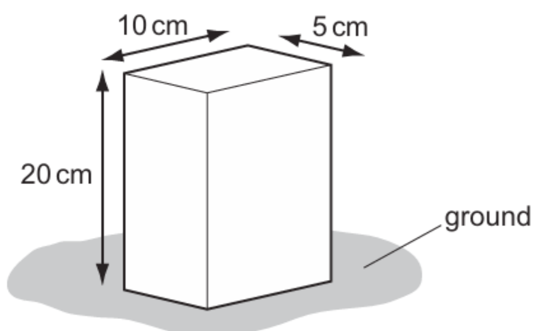
B The force on the base of S is the same as the force on the base of T.

C The pressure on the base of S is greater than the pressure on the base of T.

D The pressure on the base of S is the same as the pressure on the base of T.

43. M/J/13/11/Q8

A brick of weight 80 N stands upright on the ground as shown.



What is the pressure it exerts on the ground?

A $0.080\,N/cm^2$ **B** $0.40\,N/cm^2$ **C** $0.80\,N/cm^2$ **D** $1.6\,N/cm^2$

44. O/N/12/11/Q11

A garden table weighs 40 N and has a top surface of area $2\,m^2$. It is raining and the rain produces a pressure of $4\,N/m^2$ on the table.

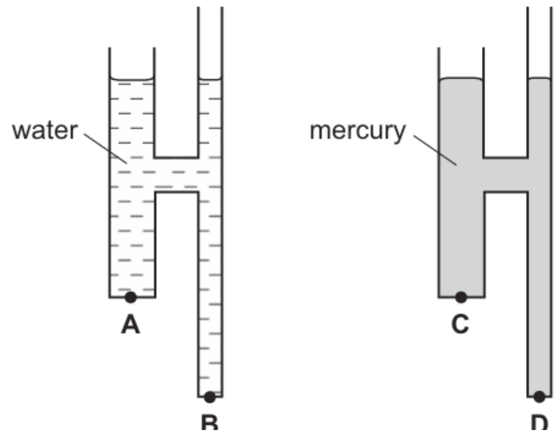
What is the force exerted by the table on the ground?

A 20 N **B** 32 N **C** 42 N **D** 48 N

45. O/N/11/11/Q13

The diagram shows two identical pieces of apparatus. One is filled with water and the other is filled with mercury. Water is less dense than mercury.

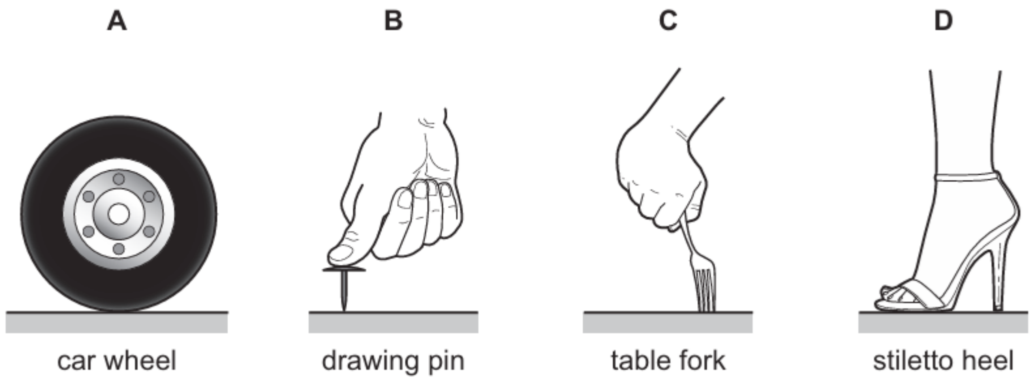
At which point is the liquid pressure greatest?



46. M/J/11/11/Q10

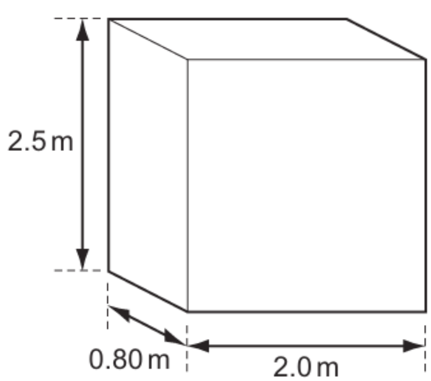
The **same** downward force is applied to four objects resting on a horizontal surface.

Which exerts the greatest pressure on the surface?



47. O/N/10/11/Q11

The base for a statue rests on level ground. It is made from stone and is 2.0 m long, 2.5 m high and 0.80 m wide. It has a weight of 96 000 N.

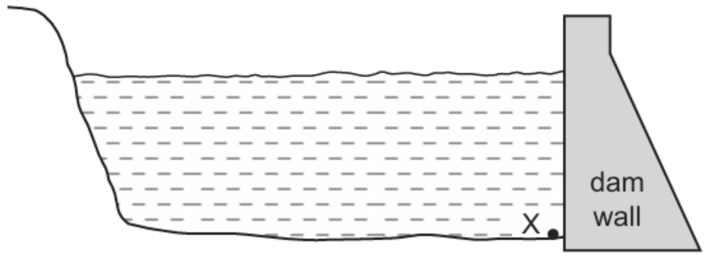


What is the pressure that the base exerts on the ground?

A 19 kPa **B** 24 kPa **C** 48 kPa **D** 60 kPa

48. O/N/10/11/Q12

An engineer designs a dam wall for a reservoir.



Which factor determines the pressure at X?

A the depth of the water in the reservoir

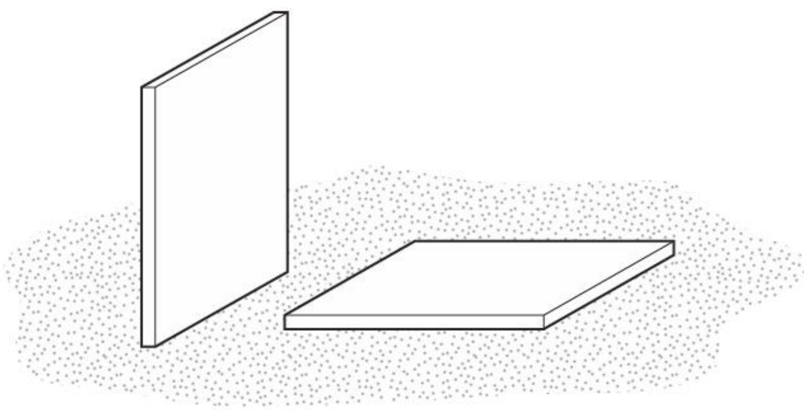
B the surface area of the reservoir

C the length of the reservoir

D the thickness of the dam wall

49. M/J/09/Q10

A builder leaves two identical, heavy, stone tiles resting on soft earth. One is vertical and the other is horizontal.



After a few hours, the vertical tile has started to sink into the soft earth, but the horizontal one has not.

Which row correctly compares the forces and the pressures that the tiles exert on the earth?

	forces	pressures
A	different	different
B	different	same
C	same	different
D	same	same

50. M/J/08/Q12

Which statement about the pressure in a column of liquid is correct?

A It acts only vertically downwards.

B It increases if the column width increases.

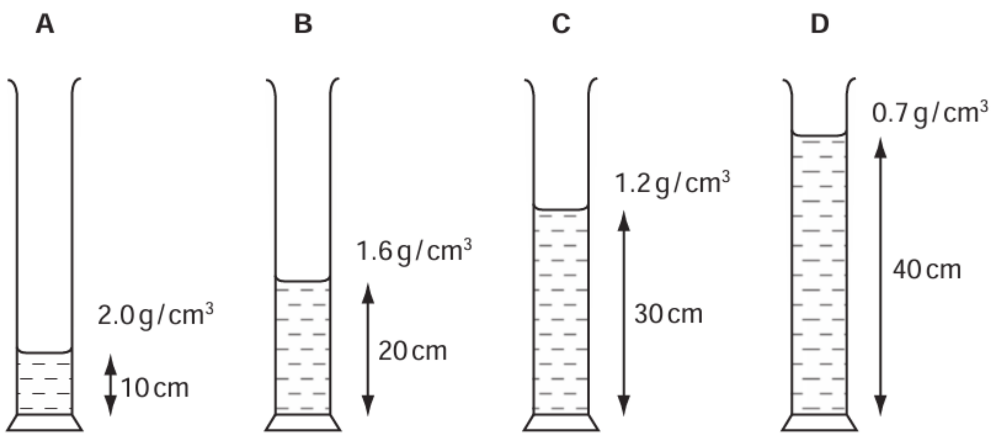
C It increases with depth in the column.

D It is uniform throughout the column.

51. O/N/07/Q12

Four different liquids are poured into identical measuring cylinders. The diagrams show the depths of the liquids and their densities.

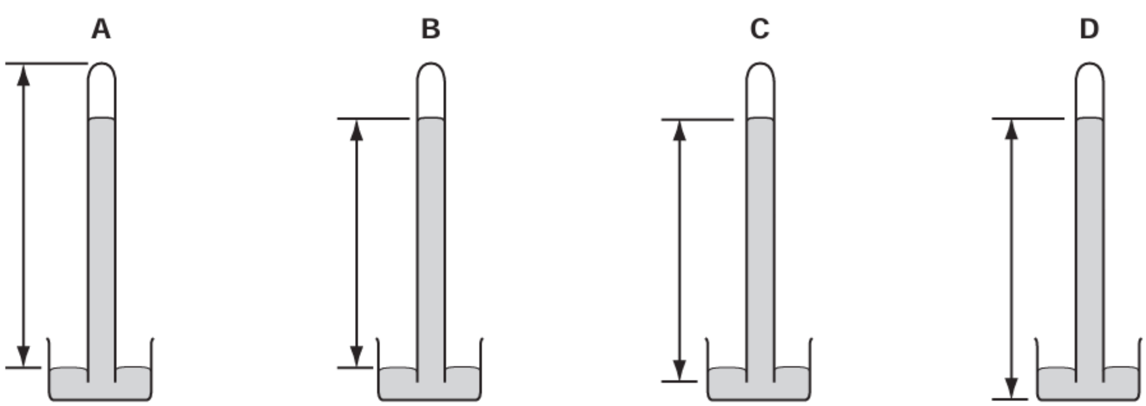
Which liquid causes the largest pressure on the base of its measuring cylinder?



52. O/N/06/Q10

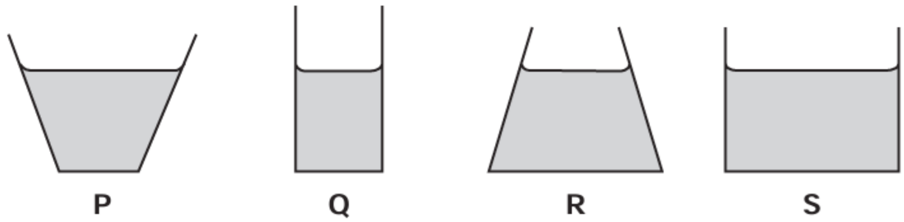
The diagrams show a simple mercury barometer.

Which diagram shows the distance to be measured to find atmospheric pressure?



53. M/J/06/Q14

The diagrams show, to the same scale, the vertical sections of a set of circular vessels. Each vessel contains the same depth of water.



Which of the following statements is correct?

A The water exerts the greatest pressure on the base of vessel **P**.

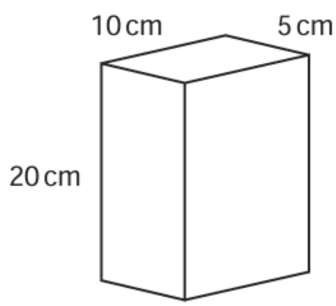
B The water exerts the greatest pressure on the base of vessel **S**.

C The water exerts the same force on the base of each vessel.

D The water exerts the same pressure on the base of each vessel.

54. O/N/05/Q13

A brick of weight 80 N stands upright on the ground as shown.



What is the pressure it exerts on the ground?

A $\frac{80}{20 \times 10}$ N / cm^2

B $\frac{20 \times 10}{80}$ N / cm^2

C $\frac{80}{10 \times 5}$ N / cm^2

D $\frac{10 \times 5}{80}$ N / cm^2

55. M/J/05/Q12

What does **not** affect the pressure at a point beneath the surface of a liquid?

A area of the liquid surface

B density of the liquid

C depth of the point below the surface

D strength of the gravitational field

56. M/J/05/Q13

A small table weighing 40 N stands on four legs, each having an area of $0.001\,m^2$.

What is the pressure of the table on the floor?

A $400\,N/m^2$ **B** $1000\,N/m^2$ **C** $10\,000\,N/m^2$ **D** $40\,000\,N/m^2$

57. O/N/04/Q14

Which would be the **least** likely to sink into soft ground?

A a loaded lorry with four wheels

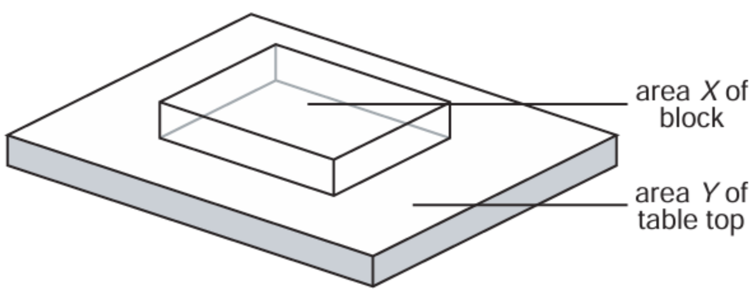
B a loaded lorry with six wheels

C an empty lorry with four wheels

D an empty lorry with six wheels

58. M/J/04/Q14

The diagram shows a glass block resting on a table top.



The area of the block in contact with the table is X and the area of the table top is Y.

The weight of the block is P and the weight of the table is Q.

Which expression gives the pressure exerted on the table by the block?

A $\dfrac{P}{X}$ **B** $\dfrac{P}{Y}$ **C** $\dfrac{Q}{X}$ **D** $\dfrac{Q}{Y}$